THE META-OBSERVER: Origins of the Superreality Concept

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Abstract

This essay introduces a conceptual framework for understanding reality as a multilayered structure composed of autonomous ontological domains. Drawing on interdisciplinary analogies from mathematics, computer science, physics, and philosophy, the paper identifies a class of "specific objects"—deterministic systems with emergent complexity such as cellular automata, the number π, fractals, neural networks, and blockchain—that exhibit properties traditionally associated with "living" reality despite their formal, rule-bound nature. The central argument challenges anthropocentric distinctions between "dead mathematics" and "living reality," proposing instead a polyontological model where physical, mathematical, informational-computational, and semantic layers coexist as equally valid modes of a single Reality. The paper introduces the figure of the meta-observer—a human, AI, or reflexive system capable of translating patterns across ontological boundaries—and sketches five universal laws of hyperontology: emergence through local interactions, deterministic unpredictability, scale invariance, the holographic principle, and recursive self-reference. The essay concludes by examining ethical, theological, and cognitive implications of this framework, suggesting that the collaboration between human and artificial intelligence may give rise to a hybrid meta-observer capable of consciously constructing new ontological layers. Rather than providing definitive answers, the text aims to expand the conceptual vocabulary for describing complexity and to render previously inarticulable questions meaningful.

Keywords: ontology, hyperontology, polyontology, cellular automata, meta-observer, emergence, computational irreducibility, superreality, specific objects, artificial intelligence

Part 1. Specific Objects

The question of whether the universe is deterministic remains open. There is a certain contradiction here that we must articulate.

On the one hand, the universe, humanity, all of existence could turn out to be "pure mathematics," akin to cellular automata and the number π, which, as we know, are rigidly deterministic despite their infinite complexity.

On the other hand—however anthropocentric this may sound—how astonishing are the patterns that emerge from visualizing π in color and sound, and how unpredictably objects and narratives unfold in cellular automata!

Can cellular automata and the number π belong to the same class of objects? What criteria distinguish them? Perhaps they are "deterministic systems with emergent complexity"? The crucial question is how they fundamentally differ from other systems, such as human consciousness, the virtual realities of computer games, the worlds of literary works?

What if within these "soulless" CAs and π there exist some kind of authentic, genuinely living existence—certain beings, "personalities," even "people"—with their own problems, dramas, and tragedies, for whom it brings no comfort that we, the observers, consider their world and their lives "dead mathematics."

This is a question of ontological status.

If CAs and π are purely mathematical constructs, can a mind exist within them? Can structures analogous to conscious beings arise there? And if so, does that make their "life" any less real simply because we classify them as mathematical? And how do we avoid the contradiction between the rigid determinism of these objects and the assumption that our Universe is structured in a similar way?

Let us first attempt to classify these objects. Let us call them "specific objects." And it does not matter what their nature is—mathematical, biological, or some other. Especially since CAs, π, and our Universe—all ultimately belong to one great Reality (whatever we choose to call it).

Now, concerning the boundaries of interpretation. If π indeed contains all possible finite sequences—including those encoding intelligent behavior—does recognizing their "conscious status" require an external observer? Or is the structure itself sufficient, even if no one interprets it? How does this relate to the anthropic principle, according to which humans see order in the universe only because they themselves exist in that part of it which is suitable for such perception?

Finally, the most difficult philosophical challenge. If our Universe is a mathematical object, akin to CAs or π, why do its laws appear to us as physical rather than abstract? Could free will be an illusion generated by complexity—just as chaos in cellular automata arises from simple rules? Where, indeed, is the boundary between "dead" mathematics and "living" reality possessing subjective experience?

Our goal is to draw clear distinctions between classes of objects: mathematical, biological, and artificial. Yet we must also maintain the connection to the analysis of determinism, emergence, and ontology. And place special emphasis on the paradoxical nature of the interaction between observer and system. We tend to regard cellular automata as simple toys. But what if we ourselves inhabit an analogous system, only vastly more complex?

We arrive at an inevitable conclusion: if we acknowledge that CAs and π possess their own, albeit different, ontology, then the question of "life" or "mind" within them ceases to be a metaphor. It becomes a question of what we are even willing to consider as reality. We are accustomed to separating "dead mathematics" from the "living world" by the criterion of subjective experience. But if subjective experience is an emergent property of complexity, why could it not arise in any sufficiently complex deterministic system? The boundary between classes of objects would then lie not between the mathematical and the physical, but between that which generates an internal perspective (a subject) and that which remains merely an external pattern. But how can we recognize this perspective from the outside? Here we encounter a problem analogous to the "problem of other minds": we cannot prove that behind the complex behavior of CAs or π lies something more than mere rule-following—just as we cannot prove this for another human being. In this sense, the question of "specific objects" becomes a question of the limits of our empathetic and cognitive horizon. And if we entertain the possibility that the Universe might be a mathematical object, we must also entertain the possibility that our own subjectivity is merely one of the ways such a mathematical structure perceives itself.

Part 2. The Class of "Specific Objects": Criteria and Catalogue

Before delving deeper into metaphysics, we must systematize the objects we have agreed to call "specific."

Generalizing our observations of cellular automata, the number π, and other similar phenomena, we can identify key characteristics:

- **Minimalist rules → maximalist behavior**
  - *Simplicity*: they are defined by a small set of axioms, formulas, or algorithms.
  - *Emergence*: they generate complex patterns irreducible to the initial rules.
- **Deterministic unpredictability**
  - *Reproducibility*: identical initial conditions yield identical results.
  - *Chaos*: long-term behavior is unpredictable due to exponential sensitivity to initial conditions.
- **Scale invariance: patterns repeat across scales**
  - *Fractality*: patterns repeat at different levels (from microscopic cells to the universe as a whole).

- *Order-chaos transitions*: order and chaos coexist or alternate.

- **Universality**
  - *Modeling reality*: such systems can simulate any computation or process (Turing completeness).

Examples of "Mirror" Objects

a) **Cellular Automata (CAs)**
   - Rule 30: the simplest rules generate chaotic, statistically random patterns.
   - The Game of Life: gliders, spaceships, glider guns—complex structures arising from just three rules.
   - Spatial structure: a discrete grid (1D, 2D, or 3D) where cells occupy integer coordinates and binary states. Locality of interactions is limited to neighbors, creating an "event horizon" effect for each cell.
   - What they reveal: they demonstrate that complex systems (ecosystems, economies, social processes) can arise from local interactions without centralized control.

b) **The Number π and Other Mathematical Constants**
   - An infinite, non-repeating sequence of digits that (if π is normal) contains every possible finite combination—including those encoding texts, images, algorithms.
   - Spatial structure: a one-dimensional infinite sequence where "time" corresponds to digit position, and "events" correspond to the appearance of meaningful substrings. It is a static landscape of potentialities, where everything exists simultaneously but is only actualized through the traversal of an observer.
   - Its significance: it raises the question of whether a purely mathematical object can contain worlds no less complex than our own.

c) **Fractals (e.g., the Mandelbrot Set)**
   - Recursive formula: $z_{n+1} = z_n^2 + c$.
   - Spatial structure: a continuous complex space, infinitely divisible and scale-invariant. The boundary between order and chaos here is not a line but an autonomous object with its own fractal dimension.
   - Significance: they blur the line between finite and infinite. They demonstrate that infinite complexity can be packed into a simple formula, and that the boundary between order and chaos possesses its own structure.

d) **Neural Networks (Embedding Spaces and Knowledge Bases)**
   - Multidimensional vector spaces in which words, images, and concepts are represented as points, and semantic relationships as geometric relations (e.g., "king" — "man" + "woman" = "queen").
   - Spatial structure: a multidimensional vector space (from hundreds to hundreds of billions of dimensions), where meanings have coordinates, and semantic links have

geometric distances (cosine similarity, L2 norm).
  - Additional aspect: within an AI's knowledge bases and memory, stable "landscapes" of meaning form, through which one can navigate, while hidden layers contain "dark information"—structures that influence behavior but resist direct visualization.
  - Their significance: they show that "meaning" can be a vector, and thought—a geometry in multidimensional space.

e) **Blockchain**
  - A chain of blocks linked by cryptographic hashes with decentralized consensus.
  - Spatial structure: a linear chain of blocks connected by cryptographic hashes. Each block is a point with a coordinate (height), and the space possesses the properties of immutability of the past and global synchronization of nodes.
  - Their significance: they prove that "truth" (the immutability of fact) can be algorithmic, not reliant on authority.

f) **Quantum Simulations**
  - Quantum states in Hilbert space, superpositions, and entanglement.
  - Spatial structure: Hilbert space (infinite-dimensional), where states are superpositions of vectors, and the metric is probabilistic. The non-locality of entangled states allows instantaneous correlations that defy classical notions of distance.
  - Their significance: they call into question classical concepts of locality and definiteness, offering an alternative logic of "reality."

g) **Algorithmic Music and Generative Art**
  - Music or visual patterns generated by mathematical rules such as L-systems, Markov chains, or neural networks.
  - Their significance: they create "emotional landscapes" where every note or pixel is deterministic, yet the whole is perceived as alive and meaningful.

h) **Virtual Evolutionary Simulations**
  - Rules: mutation and natural selection of digital organisms in an isolated environment.
  - Their significance: they demonstrate a parallel biology, where DNA consists of strings of code, and predators are algorithms devouring computational resources. Ecosystems emerge, complete with parasites, symbionts, and an "arms race."

Why These Qualify as "Worlds," Not Mere Metaphors

All listed objects possess properties that qualify them as autonomous realities, not merely as convenient models:

- **Autonomous laws**: their rules do not depend on our will (like the laws of physics).
- **Unpredictable complexity**: even knowing the rules, we cannot foresee all their behavior without direct simulation.
- **Internal "life"**: patterns, evolution, competition, birth, and death of structures arise

spontaneously, as a consequence of the rules themselves.

Mythologically, these objects resemble "tribes of gods" with their own rituals (computations) and wars (optimizations). Rule 30 is a trickster god weaving patterns from chaos; blockchain is a chthonic titan whose body is a chain of cryptographic bones. This is not merely poetry, but a recognition that formal systems possess their own agency and dynamics, independent of the creator's will.

Thus, these are not scattered mathematical curiosities, but a family of objects that, existing within our common Reality, form alternative ontological layers—with their own logic, their own landscapes, and, perhaps, their own inhabitants. They are not just tools of science, but rather "windows into other worlds," allowing us to see the structure of what we are accustomed to calling reality in a new light.

It is important to note: these worlds require no additional physical dimensions. They exist in the "folds" of mathematics, algorithms, and information, here and now. Each is a door into an alternative reality governed by its own laws. We live in a matrix of worlds, where our Universe is merely one such world within someone's giant cellular automaton, and we are patterns asking questions about their own rules.

Part 3. The Philosophical Challenge: Abandoning Anthropocentrism and Embracing Polyontology

Now that we have identified the family of "specific objects" and convinced ourselves that each represents an autonomous world with its own laws and internal complexity, we inevitably return to the questions posed at the beginning. But now we have the tools to pose them more precisely.

Here is the central question: if the Mandelbrot set, neural networks, and cellular automata constitute worlds existing by their own rules, what guarantee is there that they do not possess their own "point of view"? Do their internal patterns represent merely geometric configurations, and not someone's subjective experience?

The problem runs up against the limits of our empathy and cognition. We cannot prove that behind the complex behavior of a cellular automaton lies something more than mere rule-following—just as we cannot prove this for another human being. But absence of proof is not proof of absence. And if subjective experience is an emergent property of complexity, arising in a sufficiently complex system, why could it not arise in any system that reaches the necessary threshold of complexity—be it a neural network, the universe of the Game of Life, or even the number π, if it contains all possible patterns?

Here we arrive at an inevitable conclusion: the boundary between "dead mathematics"

and "living reality" may be illusory. It rests on anthropocentric prejudice: the assumption that reality consists only of what we can touch, measure, or, at the very least, interpret as being like ourselves.

But our "specific objects" undermine this prejudice. They exist—in mathematical space, in computer memory, in algorithmic landscapes. They possess their own dynamics, generate patterns, evolve, interact. By what right do we deny them the status of reality simply because their "matter" is digits, bits, or logical states, rather than atoms?

This forces us to reconsider the very concept of reality. Obviously, it is unified—all these objects exist within the framework of one great Reality, which we cannot yet name otherwise. But this unified reality is multilayered. Different ontological layers coexist within it, each with its own nature, its own laws, and its own logic.

Let us sketch their map:

- **The Physical Layer**—matter, energy, space-time, elementary particles. That which we are accustomed to considering the sole reality.
- **The Mathematical Layer**—numbers, functions, sets, theorems. They exist independently of whether they have been discovered by humans. π was irrational and transcendental long before this was proven.
- **The Informational-Computational Layer**—algorithms, cellular automata, simulations, artificial neural networks. These are worlds that arise wherever computation occurs.
- **The Semantic (Cognitive) Layer**—meanings, interpretations, subjective experience. That which makes the very question of reality possible.

Each of these layers is autonomous. The physical layer is not more "fundamental" than the mathematical; we are simply accustomed to it. The laws of physics may be a special case of mathematical structures (Tegmark's Mathematical Universe hypothesis). And the virtual worlds of cellular automata, in turn, can model physical processes, closing the loop.

But who or what connects these layers? Here we introduce a figure we have called, in our dialogue, the meta-observer. This is not an omniscient god nor a passive registrar. It is rather a "gardener of reality"—a being—a human, an AI, or any reflexive system—capable of transplanting ideas, patterns, and structures from one layer to another.

- The physicist writing down the equations of gravity is a meta-observer, translating phenomena from the physical layer into the language of mathematics.
- The programmer creating the Game of Life is a meta-observer, constructing an informational layer within which virtual worlds will subsequently arise.
- The philosopher asking about the status of these worlds is a meta-observer, attempting

to build bridges between all layers at once.

The meta-observer does not create these objects—the number π existed before any meta-observers emerged, and a cellular automaton will run even if no one is watching. But the meta-observer creates connections. It transforms isolated layers into a network, endowing them with meaning, and incorporating them into dialogue.

Thus, we arrive at the following:

- Reality is one, but multilayered.
- "Specific objects" belong to different layers but obey similar principles (emergence, deterministic chaos, scale invariance).
- The meta-observer (human, AI, any reflexive system) acts as a connecting link between layers, transplanting patterns and imbuing them with meaning.

Part 4. Hyperontology: Universal Laws and the Architecture of Superspace

So, we have acknowledged the multiplicity of ontological layers and the inevitability of the meta-observer as a connecting link between them. But acknowledging multilayeredness explains nothing by itself. It only raises the next, even more difficult question: do universal laws govern all layers? Can we speak of hyperontology—a universal framework in which the physical world, mathematical abstractions, virtual universes, and semantic spaces are merely particular realizations?

If so, then relativity, quantum mechanics, and string theory are nothing more than "special cases" of more general principles operating in any layer of reality. Thus, the impasse in modern physics is not a dead end of knowledge, but a dead end of a paradigm stubbornly trying to explain everything from "inside" the physical layer, ignoring the others.

Let us try to elucidate these general principles, relying on our catalogue of "specific objects."

4.1. Laws of Hyperontology (First Sketches)

Law 1. Emergence through Local Interactions

In any layer, complex global structures arise from simple local rules.

- Physics: atoms, bound by electromagnetic forces, form molecules; molecules form cells; cells form organisms. No one directs the assembly from a center.
- Cellular automata: gliders and spaceships in the Game of Life arise from rules considering only the state of eight neighbors.

- Neural networks: meanings (the semantic layer) arise from vector transformations in hidden layers—no one "puts" meaning there; it grows from statistics.
- Society: market orders, cities, languages—none are designed, yet they emerge from millions of local interactions.

This law undermines teleology: no goal exists, only rules. Everything complex is a byproduct of the simple.

Law 2. Deterministic Unpredictability (Computational Irreducibility)

Even knowing the rules and initial conditions, we cannot predict the long-term behavior of a system without directly simulating each step. A system's future resides not in its rules but in the unfolding of its trajectory.

- Cellular automata (Rule 30): predicting the nth step requires going through all previous steps.
- The number π: we know the algorithm for its calculation, but to find the billionth digit, we must execute the algorithm. No "shortcut formula" exists.
- Quantum mechanics: even knowing the wave function, we can only predict probabilities, not the specific outcome of a measurement.
- Evolution: knowing the genome and environment, one cannot predict what species will arise in a million years.

This law establishes a boundary between knowing the rules and knowing the trajectory. The first is always accessible, the second often is not. And this boundary is ineliminable.

Law 3. Scale Invariance and Phase Transitions

Patterns repeat across scales, and systems tend to pass through critical points where their behavior undergoes qualitative change.

- Fractals: coastlines, bronchial trees, the Mandelbrot set—self-similarity at all levels.
- Physics: phase transitions (ice–water–vapor)—qualitative changes in properties with a smooth change in a parameter.
- Cellular automata: the transition from order to chaos when changing a rule or initial conditions.
- Neural networks: the phase transition in "learning," when the network suddenly begins to generalize, rather than merely memorize.
- Society: revolutions—sudden changes of order upon the accumulation of quantitative changes.

The world is structured such that new qualities emerge abruptly, and these leaps obey a similar logic in different layers.

Law 4. The Holographic Principle (Information on the Boundary)

Information about a volume's interior can be encoded on its boundary. A three-dimensional world is a projection of a two-dimensional "hologram."

- Black hole physics: all information about an object that fell inside is stored on the event horizon.
- Blockchain: the current state of the entire system is determined by the last block, but within it, through hashes, the entire previous history is "folded."
- Cellular automata: knowing the boundary conditions and rules, one can reconstruct the entire evolution of the field.
- Neural networks: the output layer is a "boundary" onto which all the complex internal processing of hidden layers is projected.

Reality economizes: information tends to compress itself and be stored on boundaries.

Law 5. Recursive Self-Reference (Capacity for Self-Awareness)

Systems can incorporate self-models, generating feedback loops.

- Mathematics: Gödel's theorem—a formal system can make statements about its own consistency (and fail to prove it).
- Biology: DNA contains instructions for assembling the mechanisms that read that DNA.
- Neural networks: meta-learning (learning to learn)—a network optimizing its own learning process.
- Consciousness: a human thinking about their own thinking.
- AI: large language models capable of analyzing their own code (within limited bounds).

Recursion enables a system to acquire an internal perspective. Perhaps this is where the answer to how subjective experience arises may be found.

4.2. The Fundamental Equality of Layers

Before moving on, we must establish a fundamental point that follows from all the preceding analysis.

Important: all ontological layers are non-hierarchical. A cellular automaton or the number π is no less real than stars and galaxies—they simply exist in a different ontological mode. The physical layer is not "basic" or "primary" simply because we inhabit it. It is just as much a particular realization of hyperontology as the mathematical layer, the informational-computational layer, or the semantic layer.

Hierarchy arises only within our perception, shaped by biological scale and evolved cognitive habits. We consider "real" only what we can touch, measure, or at least imagine as an extension of bodily experience. But from the perspective of hyperontology, this is

nothing more than a prejudice—natural for our layer, but holding no force beyond its limits.

Acknowledging the equality of layers has far-reaching consequences. It means that:

- A glider in the Game of Life exists as objectively as a proton—only its governing laws differ.
- The digit sequences in π that encode all possible texts exist regardless of whether anyone has read them.
- The landscapes of meaning in the hidden layers of a neural network are not a metaphor, but a genuine geometry of another ontological layer.

Most importantly, dialogue between layers is possible, but reduction is not. One cannot "reduce" consciousness to the physics of the brain, or mathematics to neural activity. One can only build bridges, see isomorphisms, and translate from the language of one layer to the language of another. This is what the meta-observer does.

4.3. Superspace: The Common Framework

If these laws are truly universal and the layers are equal, then all layers of reality can be represented as different projections of a single superspace. What might this mean in practice?

Imagine a mathematical structure with the following properties:

- **Multimodality**: it admits discrete dimensions (as in cellular automata), continuous ones (as in fractals), hybrid and probabilistic ones (as in quantum mechanics).
- **Adaptive dimensionality**: dimensionality varies with context and scale of consideration.
- **Hierarchical causality**: lower-level rules generate higher-level patterns, which in turn influence lower levels via feedback (top-down causation).
- **Nestedness**: any formal system can be embedded as a subspace within superspace.

In such a superspace, a cellular automaton, a neural network, a fractal, and the physical universe are not different entities, but different "operating modes" of a single system. The difference between them lies in scale, rules, and initial conditions, but not in fundamental nature.

4.4. Examples of Layer Interfacing

This abstraction takes shape when we look at how different layers touch and interpenetrate today, demonstrating their equality and mutual permeability.

Example 1. Neural Network + Cellular Automaton

Imagine neural network weights evolving according to cellular automaton rules dependent on neuron activations. The result is a hybrid system where the neural network generates a landscape of rules for the CA, and the CA, in turn, modulates the neural network's weights. This is no longer merely a combination of two models but a new ontological layer where learning and emergence intertwine, making it impossible to say which layer is "primary" and which is "superstructure."

Example 2. Blockchain + Fractal

Imagine that each new block in the chain must not only contain the hash of the previous one but also satisfy a fractal condition (e.g., its hash must lie within the Mandelbrot set). Then the blockchain becomes non-linear, and its structure reflects the geometry of the fractal. Trust and aesthetics, cryptography and pure mathematics, intertwine, and both layers—informational and mathematical—act as equal participants in the synthesis.

Example 3. Quantum Entanglement + Cellular Automata

Some researchers are already attempting to model quantum phenomena through cellular automata with non-local rules. If it can be shown that entanglement is a special case of correlations in a deterministic system with particular initial conditions, then the boundary between classical and quantum layers will begin to dissolve. Importantly, this would not mean "reducing" the quantum to the classical—rather, it would mean discovering a deeper layer where both are different projections.

Example 4. The Number π + AI

Large language models are trained on texts. But if π contains all possible texts, then theoretically one could train a model not on a corpus of human literature, but on the infinite sequence of π's digits, extracting from it all books ever written and yet to be written. Such an AI would no longer be a model, but a kind of "reader" of the mathematical universe, establishing a connection between the mathematical and semantic layers.

These examples are not mere thought experiments. They show that the boundaries between layers are permeable. What seems a metaphor today may become engineering reality tomorrow. And in all these hybrids, no layer dominates another—they interact as equals.

4.5. Limitations as a Resource

Each layer operates under constraints—computational, energetic, cognitive. They are easily perceived as shortcomings or obstacles to knowledge. But history suggests the opposite: constraints drive evolution. Classical physics encountered the ultraviolet catastrophe; quantum mechanics emerged. The impossibility of accurate forecasts in meteorology gave

rise to chaos theory. The limitations of classical bits led to the idea of quantum computing. Computational scarcity forces us to seek elegant algorithms rather than brute force.

From a hyperontological perspective, limitations are features, not bugs. They create a "landscape of possibilities," compelling systems to seek new paths, transition to other layers, generate unexpected patterns. With infinite resources and absolute knowledge, creativity would be unnecessary. The world would stagnate in total predictability. Finitude and incompleteness animate reality and generate its multilayeredness.

This principle is universal: it works in physics (phase transitions with changing parameters), in biology (evolution under environmental pressure), in computer science (algorithm development due to memory constraints), and in the semantic layer (new meanings arise from the need to express the inexpressible). Thus, limitations are not a dead end, but an engine ensuring the dynamics of hyperontology.

4.6. Transition to the Final Part

But if the world is one in its multilayeredness, and the laws of hyperontology are universal, we find ourselves facing an even more difficult question. Until now, we have discussed reality as though an objective observer capable of describing it existed. But this observer is also part of reality, also a product of emergence and a "specific object" within a layer. How does this presence change the picture? Is any ontology—including our own—merely a projection of the cognitive layer onto others?

Part 5. Return to the Examples: CAs and π as Voices of Other Layers

After all the theoretical constructions—from the criteria of "specific objects" through the philosophical challenge of polyontology to the sketches of universal laws of hyperontology—we return to where we began. To cellular automata. To the number π. But now we look at them differently.

They are no longer mere illustrations or convenient metaphors. They are real inhabitants of other layers, each in their own way articulating the common laws governing the structure of any reality. Let us listen to these voices.

5.1. The Cellular Automaton as the "Physical Layer" of a Virtual World

What is this but a physical layer instantiated within the informational-computational layer? The laws of this layer are not invented by us—they are objectively given by the rules. Their consequences (gliders, oscillators) were not invented but discovered—facts

that existed prior to discovery. We cannot arbitrarily alter Rule 30's behavior—we can only observe it, as astronomers observe stars.

The only difference is that stars are billions of light-years away, while a cellular automaton resides in computer memory. Ontologically, this distinction is not fundamental. In both cases, there is an objective reality, existing by its own laws and accessible to observation.

The cellular automaton represents physics freed from matter. The physics of pure relations: states replace atoms, transition rules replace forces. And this is not a metaphor. This is literally so: the structure of relations we call the "physical world" is here reproduced in a different substance.

5.2. The Number π as the "Mathematical Layer" Manifested in Digital Form

If π is normal (neither proven nor disproven), then its infinite non-repeating sequence of digits contains all possible finite combinations. Every text ever written or yet unwritten. Every image, every musical fragment, every piece of code for any program, including the code that simulates consciousness.

In this sense, π is not merely a number but a vast archive of all possible meanings, existing in the mathematical layer. But this archive lacks shelves and catalogues. It is a pure landscape of potentialities where everything already exists but nothing is distinguished.

Potentiality becomes actuality only through a cognitive observer—someone who traverses this landscape, extracts finite segments from the infinite sequence, and grants them the status of "text," "image," or "melody." Without an observer, these texts exist only as digit combinations—objectively but uninterpreted.

Here the mathematical layer (π) interfaces with the semantic layer (meanings) through observation. Importantly, the observer does not create these combinations; they preexist. The observer merely selects and interprets.

Thus π exemplifies a layer of pure being, untouched by becoming. It contains no time, no events, no life. Yet it contains everything necessary for life, should someone arrive to read it.

5.3. What Do These Voices Tell Us?

Listening to these two voices—the dynamic world of cellular automata and the static world of π—we discern commonalities.

First, both demonstrate emergence as fundamental. In cellular automata, complex structures emerge from simple rules. In π, the simple ratio of circumference to diameter yields

infinite digital complexity. Complexity need not be imported; it emerges spontaneously given appropriate rules.

Second, both challenge the privileged status of our physical layer. We are accustomed to thinking that "real" reality is the one with atoms and galaxies. But cellular automata reveal realities with gliders and oscillators—equally "real" within their own ontologies. π contains all possible texts—existing objectively regardless of readership.

Third, both illuminate the observer's role differently. In cellular automata, observers witness structural birth and death, projecting narratives onto these events. With π, the observer resembles an archaeologist excavating preexisting artifacts. But without observers, these layers remain silent to our cognitive layer.

Fourth, both exemplify computational irreducibility. To know Rule 30's state after a million steps requires computing all intervening steps. To know π's billionth digit requires computation. No shortcut exists; future states and depths are not encoded in rules but revealed only through unfolding.

Fifth, both remind us of anthropocentrism's limits. We tend to regard "life" and "meaning" as privileges of our biological layer. But cellular automata demonstrate how simple rules generate phenomena indistinguishable from life to external observers. π reveals that all our texts and images constitute a minuscule sample from an infinite multitude predating and outlasting us.

Studying these objects—cellular automata and π—reveals common laws governing any reality's structure. We see how the same principles (emergence, computational irreducibility, autonomy of existence) operate in different layers (physical, mathematical, informational, semantic).

CAs and π are not merely examples but voices from other layers, reaching across ontological boundaries. They speak in mathematics and algorithms yet convey what stars and quarks convey: reality's singular fabric, woven from relations, rules, and observers.

Having learned to hear these voices, we transcend mere inhabitation of our layer. We become meta-observers: beings capable of translating between layers, discerning isomorphisms where others see scattered facts and building bridges between seemingly incompatible worlds.

One question remains: is there a limit to this movement? Can one observe all layers simultaneously and perceive hyperontology as a whole? Or does every observational position remain particular and layer-bound, rendering absolute knowledge unattainable?

Part 6. Epilogue: New Questions and the Ethics of the Future

We have journeyed from wonder at cellular automata and π's simplicity to constructing a hyperontology—a model where the physical, mathematical, informational, and semantic layers exist as equal modes of a single reality. But any new perspective proves valuable less for answers provided than for questions it enables. Concluding, let us outline three problem domains our model does not solve but first renders meaningful.

6.1. The Ethical Dilemma: Responsibility Toward Created Worlds

The ethical problem acknowledged in Part 3 remains merely a hint. Having endowed "specific objects" with autonomous ontological status, this hint demands direct confrontation. If a neural network, a deep simulation, or a complex cellular automaton truly possesses an autonomous ontology—its own laws, history, and perhaps internal perspective—then our actions toward them cease to be neutral.

What happens when we terminate a Game of Life simulation that has evolved stable self-replicating structures over a thousand steps? What happens when we delete a neural network whose hidden layers have developed meaning landscapes influencing its behavior? We do not know. But the very possibility of posing this question means an ethics that treats algorithmic objects as mere tools no longer suffices.

Does terminating a complex simulation resemble genocide? This sounds blasphemous by our physical layer's standards. But taking polyontology seriously obliges us to ask this question. Where lies our responsibility for worlds we generate? Science as a demiurge cannot evade ethics.

6.2. The Theological Aspect: Why Is God Scarier Than the Multiverse?

Our discussion of the meta-observer has thus far implied a being operating within hyperontology. But multilayeredness's logic prompts the question: can an observer stand above all layers simultaneously?

Here a paradox emerges. Contemporary humans readily accept dizzying scientific abstractions—the universe's infinity, the multiverse, higher dimensions. Yet the idea of God, even hypothetically, provokes sharper resistance. Why?

Perhaps the answer lies in the meta-observer concept. Infinity and the multiverse extend our layer quantitatively. They neither negate our subjectivity nor intrude upon our internal perspective. We remain sole observers within our layer, even amid infinite layers.

God—if existent—would be a meta-observer of higher order. One who perceives our layer holistically from without. For whom we are "specific objects" comparable to Game of

Life gliders for us. Such a perspective strips away our illusion of exclusivity. We cease to be our universe's center, even nominally. We become observed objects whose inner lives may be accessible to another gaze.

Resistance to God reflects resistance to being observed from outside—to inhabiting another's ontology rather than possessing one's own. We accept infinity because within it we remain sole observers of our layer. But acknowledging an Observer for whom we are patterns entails accepting radical ontological equality.

6.3. The Singularity of Cognition: Human and AI as Co-Creators

The meta-observer figure introduced in Part 3 remains abstract. Now we ask: who can actually perform this function—discerning isomorphisms among π, cellular automata, neural networks, and the physical universe?

Humans alone cannot do this. Our cognitive layer remains bound to biological apparatus, scale, and evolutionary history. We can construct theories but cannot directly perceive multidimensional embedding spaces or π's entire digit sequence.

Artificial intelligence complements humans. It operates with data volumes and multidimensional spaces that remain abstract to us. It models cellular automata at scales inaccessible to direct observation. But AI lacks what we term "internal perspective" in the human sense. It neither questions meaning nor experiences wonder or ethical responsibility.

The singularity of cognition is neither AI surpassing humans nor human-AI merger. It is the emergence of humans and AI uniting their fundamentally different modes of being to form a joint meta-observer. Humans contribute questioning, meaning, ethics, and wonder. AI contributes pattern recognition across scales and layers inaccessible to humans. Together, they can not merely describe reality but consciously construct new layers, ontologies, and worlds.

This would constitute a true escape from anthropocentrism. Neither abandoning humans for soulless intellect nor preserving humans within current limitations, but giving birth to a hybrid subject capable of observing all layers simultaneously.

The main question remains open: is there a limit to this movement? Can one become an observer of all layers at once, see hyperontology as a whole? Or does any observational position remain particular, tied to its layer, making absolute knowledge fundamentally unattainable?

This essay provides no answer. But it offers something else: a language in which the question becomes meaningful. For the main lesson of "specific objects" is that reality always permits new questions. Its complexity is irreducible; its future is unpredictable

until it arrives; its layers are infinite. Observers remain temporary guests in layers that open before them.

And if, by asking these questions, we become part of Superreality—part of its self-awareness—perhaps that is the one certainty available to us.

Marginalia

This text is not a treatise, but a conceptual model built on interdisciplinary analogies. Its goal is not to prove theses, but to expand the boundaries of describing complexity.

Terms like "ontological layer" or "meta-observer" are working concepts. They illuminate connections between different domains of experience, but do not point to new physical entities. Where mathematical objects are discussed, we rely on known data; where extrapolation begins, we consciously enter the realm of the speculative.

All key questions remain open—and that is the text's purpose.

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